

OC Core 1.3 / Mathematics Domain Packet

Status Snapshot

- Current state: `VALIDATED_ANCHOR_ACTIVE`
- Target state: `VALIDATED_ANCHOR_ACTIVE`
- Scientific class: `THEOREM_NATIVE`
- Trace status: `TRACE_COMPLETE`
- Closure verdict: `PASS`
- Quantitative pass result: `PASS_TERMINALIZED_NUMERICAL_REPLAY`
- Next required action: `MAINTAIN_REPLAY_DISCIPLINE`
- Replay status: `PASS_REPLAYABLE`

Theorem-to-Observable Map

Benchmark Dataset Manifest

Pinned Official Route Protocol

- `MATH_ROUTE_001`: Clay Mathematics Institute (`OFFICIAL_PROBLEM_CATALOG`) -
> <https://www.claymath.org/millennium-problems/> | purpose: Exact benchmark anchor catalog
- `MATH_ROUTE_002`: American Institute of Mathematics (`OFFICIAL_PROBLEM_CATALOG`) -
> <https://aimath.org/problemlists/> | purpose: Supplementary exact-question corpus

Measurable Outputs

- `epsilon_star`
- `minimal_denominator`
- `minimal_augmentation_rank`
- `component_fiber_count`
- `move_connectivity_or_degree_bound`
- `strict_terminalized_total`
- `counterexample_executed_total`
- `python_evidence_executed_total`

Measurement Schema

Replay declared mathematical packets deterministically against the strict closure ledger and counterexample traces.

Acceptance Criterion

No replay divergence and no surviving counterexample against the promoted packet.

Falsifier

A single replay divergence or surviving counterexample collapses promotion for the affected packet.

Quantitative Acceptance Thresholds

- counterexample_survivor_total_max: 0
- policy_snapshot: {'benchmark_residual_policy': {'critical_residual_sigma_threshold': 2.5, 'schema_id': 'LOGION_BENCHMARK_RESIDUAL_POLICY_v1', 'severe_residual_sigma_threshold': 1.5}, 'prediction_accuracy_policy': {'brier_score_max': 0.08, 'cases_total_required': 30, 'coverage_ratio_required': 1.0, 'critical_failure_f1_min': 0.87, 'critical_failure_precision_min': 0.85, 'critical_failure_recall_min': 0.9, 'expected_calibration_error_max': 0.05, 'schema_id': 'LOGION_PREDICTION_ACCURACY_POLICY_v2'}, 'prediction_sigma_policy': {'cases_total_min': 30, 'normalized_error_max_max_sigma': 5.0, 'normalized_error_mean_abs_max_sigma': 1.0, 'normalized_error_p95_max_sigma': 2.5, 'schema_id': 'LOGION_PREDICTION_SIGMA_POLICY_v1', 'tail_breach_count_max': 0}}
- replay_divergence_total_max: 0
- strict_terminalized_total_min: 31

Replay Harness

- Replay command: `simulations/run_all.py`
- Evidence bar: `HYBRID_ESCALATION`
- Protocol id: `OC13::EXECUTION::MATHEMATICS::ANCHOR_REPLAY`

Institute-run Escalation

- Current escalation status: `NOT_REQUIRED`
- Escalation trigger: `ONLY_IF_DETERMINISTIC_REPLAY_BREAKS`
- Measurement wave id: “
- Measurement plan: Expand the exact-question corpus instead of broadening claims.

Blocking IDs

- None.